MIL-C-492A
6 MARCH 1959
SUPERSEDING
JAN–B–492
28 AUGUST 1947

## MILITARY SPECIFICATION

# CART, LUMBER, TWO-WHEEL

*This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.*

### 1. SCOPE

1.1 This specification covers a 2-wheel lumber cart.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

**SPECIFICATIONS**

FEDERAL

QQ–S–781 — Strapping, Flat; Steel.

PPP–T–60 — Tape; Pressure-Sensitive Adhesive, Waterproof — For Packaging and Sealing.

MILITARY

MIL–P–116 — Preservation, Methods of.

MIL–B–121 — Barrier Material, Greaseproofed, Flexible (Waterproofed).

MIL–C–132 — Crates, Open, Wood; Maximum Capacity 2,500 Pounds.

MIL–P–514 — Plate, Identification and Plate, Identification Transportation-Data.

MIL–T–704 — Treatment and Painting of Materiel.

MIL–G–3278 — Grease, Aircraft and Instrument (for Low and High Temperature).

MIL–F–3541 — Fittings, Lubrication (Hydraulic).

MIL–G–3859 — Grease Guns, Hand, High Pressure, Lever-Operated (15-ounce and 21-ounce cap).

MIL–G–10924 — Grease, Automotive and Artillery.

MIL–C–11456 — Crates; Wood, Nailed, Unsheathed: Domestic Shipment, 1,500 Pounds Maximum Load.

FSC 3920

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## STANDARDS

MILITARY

MIL–STD–129 — Marking for Shipment and Storage.

MIL–STD–130 — Identification Marking of U. S. Military Property.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

**2.2 Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules.

(Application for copies of the above publication should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules.

(Application for copies of the above publication should be addressed to the American Trucking Association, 1424 16th Street NW, Washington, D. C.)

## 3. REQUIREMENTS

**3.1 Description.** The lumber cart shall be of the 2-wheel type and shall consist of a steel wheel-and-axle assembly attached to a wooden frame (on which the lumber is to be carried) and a steel leg for holding the frame approximately horizontal when the cart is stationary.

**3.2 Material.** Material shall be as specified herein.

**3.2.1** *Lumber.* Wood parts shall be of any of the following: ash, beech, yellow birch, hickory, hard maple, oak, or pecan. Moisture content shall not exceed 18 percent. Slope of grain shall not exceed one in 10 on any face. Wood parts shall be free of decay, splits, warp, and unsound knots. Other defects shall be limited as follows:

(a) Wane shall not exceed ⅛ the width of face in which it occurs.

(b) Holes shall not exceed ½ inch in diameter and shall be not less than 6 inches apart.

(c) Sound, tight knots or knot clusters shall not exceed ¼ the width of the face in which they occur.

(d) Surface checks and shake shall not exceed 1/32 inch in width and 10 inches in length.

**3.3 Dimensions and weight.** The surface of the frame on which the lumber is carried shall be 65 inches plus or minus 1 inch long by 37 inches plus or minus 1 inch wide, and shall be 27 inches plus or minus 1 inch above the ground. The weight of the cart shall not exceed 425 pounds.

**3.3.1** *Capacity.* The cart shall have a rated load capacity of 4,000 pounds.

**3.4 Wheel-and-axle assembly.** The wheels shall be of all-steel construction, shall be 36 inches plus or minus ½ inch in diameter, and shall have steel tires ½ inch plus or minus 1/16 inch thick and 4 inches plus or minus ¼ inch wide. The wheels shall be fitted with roller bearings which shall be protected against the entrance of dirt. The axle shall be fitted with flange collars to prevent axial movement of the wheels, and the wheels shall be held on with cotter pins.

**3.5 Frame.** The frame shall be provided with not less than three cross members, one in the center and one at each end. Frame members shall be securely bolted together, and metal parts attached to the frame shall be securely bolted in place. The cross mem-



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bes at the ends of the frame shall extend approximately 5 inches beyond the main body of the frame and shall have rounded edges for hand-holds. A wheel guard, consisting of a flat steel bar approximately 2½ inches wide and ⅝ inch thick, shall be securely attached to the center cross member in such a manner to allow for disassembly in accordance with the requirements of 5.1.1.1. The guard shall extend up and over each wheel to within approximately 1 inch of the outer edges of each wheel tire, and shall have a radial wheel clearance of approximately 4 inches and an axial wheel clearance of approximately 1½ inches. Steel stake pockets to accommodate 2-inch by 3-inch wood stakes shall be attached to the outer face of each side member of the frame near each corner. The stakes shall not be furnished under this specification.

3.5.1 *Support leg.* The inverted-tripod-type support leg shall be of 1½-inch by ½-inch steel and shall cear the ground by approximately 2 inches when the carrying surface is horizontal.

**3.6 Lubrication.** All moving parts shall be provided with suitable means of lubrication.

3.6.1 *Lubricants.* The cart shall be designed to operate efficiently and satisfactorily when lubricated with standard Military lubricants.

3.6.2 *Lubrication fittings.* Lubrication fittings shall conform to Specification MIL–F–3541, types I, II, and III, and shall be located so as to be accessible for lubrication by a hand gun conforming to Specification MIL–G–3859, size II.

3.6.3 *Initial lubrication service.* Initial grease lubrication, including lubricaton of sealed bearings, shall be with standard Military thermally stable greases conforming to Specifications MIL–G–10924 or MIL–G–3278 as applicable. Unless otherwise specified (see 5.1), the equipment shall be serviced with winter grades of standard Military oils d signated for use in the temperature range 0°F. to plus 32°F. before delivery to t Government. After lubrication service, ea item of equipment shall be tagged in a co spicuous place to indicate the temperatu range and grades of oils and greases used.

**3.7 Identification marking.** Each cart sha be branded "U.S." in accordance with Stan ard MIL–STD–130.

**3.8 Transportation-data plate.** A transpo tation-data plate conforming to Specificatio MIL–P–514 shall be permanently affixed t the unit in a conspicuous place.

**3.9 Treatment and painting.** All parts o the equipment normally painted in good commercial practice shall be treated and painted in accordance with Specification MIL–T–704, type A.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1** Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

**4.2 Classification of inspection.** Inspection shall be classified as follows:

(a) Acceptance inspection of the cart.

(b) Acceptance inspection of preparation for delivery.

**4.3 Acceptance inspection.**

4.3.1 *Examination.* Each cart shall be examined to determine conformance to the requirements of this specification in general



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and specifically as outlined below. Failure to pass any examination shall be cause for rejection.

(a) Dimensional.

1. Dimensions in accordance with paragraphs 3.2.1, 3.3, 3.4, and 3.5.

(b) Functional.

1. Presence of all parts and components.
2. Completeness and correctness of assembly.
3. Function of parts and components.
4. Safety devices and guards.
5. Location of lubricating devices (see 3.6.2).

(c) Materials.

1. Suitability and thickness of material.
2. Proper specified of lumber used.

(d) Workmanship.

1. Treatment and painting. Incomplete preparation before painting; such defects as scale, lumps, runs, craze, thin or bare spots; color of dry film not as specified or streaked and uneven.
2. Suitability of fit at joints, seams, edges; and misalinement.
3. Appearance of welds (porosity, flux inclusions, excessive spatter, undercutting, lack of fusion with parent metal or between parts).
4. Completeness and adequacy of fasteners.
5. Clean components.

(e) Presence and acceptability of accessories, identification and other required markings.

**4.3.2** *Test.* Each coat shall be tested as follows. Failure to pass the test shall be cause for rejection.

**4.3.2.1** *Load test.* The cart shall be test loaded with lumber at least 8 feet long weighing at least 4,000 pounds; structural beams may be substituted for lumber. Distribute lumber or beams evenly so that cart can be raised or support leg can be lifted from ground by one man. Pull lumber cart at a speed not less than 5 miles per hour about 100 yards over gravel surface having several small depressions to create shock and twisting action, after which examine frame, wheels, and bearings to assure that there has been no failure of parts due to defective materials and construction. Examine leg support with cart loaded with lumber or beams as specified above to assure that no excessive distortion exists when under load.

**4.4 Preparation for delivery inspection.**

**4.4.1** *Examination.* The preservation, packaging, packing, and marking of the cart shall be examined to determine compliance with the requirements of section 5 of this specification.

**4.4.2** *Tests.*

**4.4.2.1** *Pilot pack test.* One of the following tests shall be made. The railroad-car-method shall be used whenever such facilities are available. The crane and drop methods are alternate methods and shall be used in that order of preference only when facilities for the railroad-car method are not available.

**4.4.2.1.1** *Railroad-car method.* The crate shall be securely blocked to prevent movement on the car. Anti-skid plates shall not be



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used for this test. The car shall be humped or shunted to cause impact at approximately 10 miles per hour.

4.4.2.1.2 *Crane method.* The crate shall be raised with a crane and slings or grabhooks to a height sufficiently to allow the pack to swing clear of the ground. The crate shall be swung, causing one end or corner to strike against a solid barrier at a speed of approximately 8 miles per hour. This can be done by pulling the crate back and up along the arc of free swing until the center of balance reaches a point approximately 27 inches above its original position, also point of impact, measured vertically. The test shall be repeated for the opposite end of the crate.

4.4.2.1.3 *Drop method.* One end of the crate shall be mounted on a wood sill 5 inches or 6 inches high, placed at right angles to the skids. The opposite end shall then be raised to a height of 36 inches and allowed to fall freely to a hard surface. This test shall be applied once to each end of the crate.

4.4.2.1.4 *Examination, redesign, and retest.* After the pilot pack test, the pack shall be examined for conformance to 5.2.2.1. Any deficiencies shall be corrected by redesign, and the pack shall be retested.

## 5. PREPARATION FOR DELIVERY

### 5.1 Preservation and packaging.

5.1.1 *Level A.*

5.1.1.1 *Disassembly.* The wheels, wheel guards, and leg shall be removed. The flange collars and cotter pins shall be replaced on the axle.

5.1.1.2 *Preservatives.* Metal surfaces of the carts requiring the application of a contact preservative in accordance with Specification MIL–P–116, and as specified herein shall be coated with preservatives conforming to the applicable specifications listed in and shall be applied in accordance with Specification MIL–P–116.

5.1.1.3 *Miscellaneous components.* Unpainted exterior metal surfaces of component parts of the carts shall be coated with type P–1 preservative.

5.1.1.4 *Wheels, axles, and bearings.* The axles and flange collars shall be coated with type P–11 preservative. Bearings and interior surfaces of the hubs shall be coated with type P–11 preservative. The bearings shall be placed in the hubs, and the axles shall be wrapped and the hubs shall be covered with greaseproof barrier material conforming to Specification MIL–B–121, type I, grade C, class 1, secured in place with waterproof pressure-sensitive adhesive tape conforming to Specification PPP–T–60, type III, class 1.

5.1.2 *Level C.* The carts shall be preserved and packaged in accordance with the supplier's standard practice.

### 5.2 Packing.

5.2.1 *Pilot pack.* Before a number of carts are packed, level A, a pilot pack shall be submitted to determine conformance to the packing requirements.

5.2.2 *Level A.* Unless otherwise specified (see 6.2), the component parts comprising one complete cart shall be packed in an open wood crate conforming to Specification MIL–C–132, style B. The wheels, wheel guards, and leg for each cart shall be placed on top of the wood frame and secured in place with flat steel strapping conforming to Specification QQ–S–781, type II, class B. Blocking, anchoring, and waterproofing of the contents and strapping of the crate shall be in accordance with the appendix to the crate specification. Strapping shall be zinc coated. When specified (see 6.2), two or more carts shall be packed in a crate not to exceed the dimension and weight limitations of the crate



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specification. Blocking, anchoring, and waterproofing of the contents shall be accomplished as specified in this paragraph.

5.2.2.1 *Performance.* The packing of the carts for level A shall be capable of withstanding one of the tests described in 4.5.2.1 without shifting of contents or loosening or breaking of hold-downs, ties, stays, blocking, or bracing.

5.2.3 *Level B.* The component parts comprising one or more complete carts shall be packed in unsheathed, nailed wood crates conforming to Specification MIL–C–11456, type B–5, in quantities not to exceed the dimensions and weight limitations of the crate specification.

5.2.4 *Level C.* The carts shall be packed in containers complying with Uniform Freight Classification Rules or with National Motor Freight Classification Rules or with other carrier rules applicable to the mode of transportation.

**5.3 Marking.** In addition to any special marking required, packages and shipping containers shall be marked in accordance with standard MIL–STD–129.

## 6. NOTES

**6.1 Intended use.** The lumber cart is intended for use in transferring lumber from sawmill to stockpiles and at depots for transferring lumber from stockpiles to trucks.

**6.2 Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Administrative provisions for inspection records (see 4.1).

(c) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

(d) When two or more carts shall be packed together (see 5.2.2).

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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# SPECIFICATION ANALYSIS SHEET

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**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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